UNIT - II

2 Internet of Things : Concepts

Syllabus

Introduction to Internet of Things (IoT) : Definition, Characteristics of IoT, Vision, Trends in Adoption of IoT, IoT Devices, IoT Devices Vs Computers, Societal Benefits of IoT, Technical Building Blocks. **Physical Design of IoT :** Things in IoT, Interoperability of IoT Devices, Sensors and Actuators, Need of Analog / Digital Conversion. **Logical Design of IoT :** IoT functional blocks, IoT enabling technologies, IoT levels and deployment templates, Applications in IoT.

Contents

2.1 Introduction of Internet of Things

SPPU : May-18, Dec.-19, March-20

- The Internet of Things (IoT) refers to the capability of everyday devices to connect to other devices and people through the existing Internet infrastructure. Devices connect and communicate in many ways.
- Examples of this are smartphones that interact with other smartphones, vehicle-to-vehicle communication, connected video cameras, and connected medical devices.
- They are able to communicate with consumers, collect and transmit data to companies, and compile large amounts of data for third parties.
- Things are objects of the physical world (physical things) or of the information world (virtual world) which are capable of being identified and integrated into communication networks. Things have associated information, which can be static and dynamic.
- Physical things exist in the physical world and are capable of being sensed, actuated and connected. Examples of physical things include the surrounding environment, industrial robots, goods and electrical equipment.
- Virtual things exist in the information world and are capable of being stored, processed and accessed. Examples of virtual things include multimedia content and application software.
- A device is a piece of equipment with the mandatory capabilities of communication and optional capabilities of sensing, actuation, data capture, data storage and data processing.
- The devices collect various kinds of information and provide it to the information and communication networks for further processing. Some devices also execute operations based on information received from the information and communication networks.
- Fig. 2.1.1 shows evolutionary phase of internet.

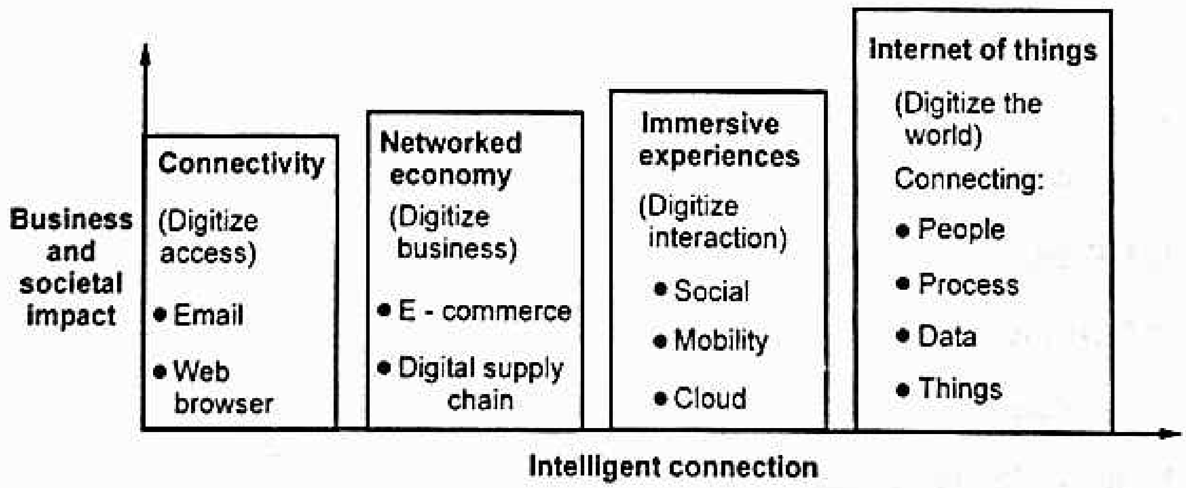
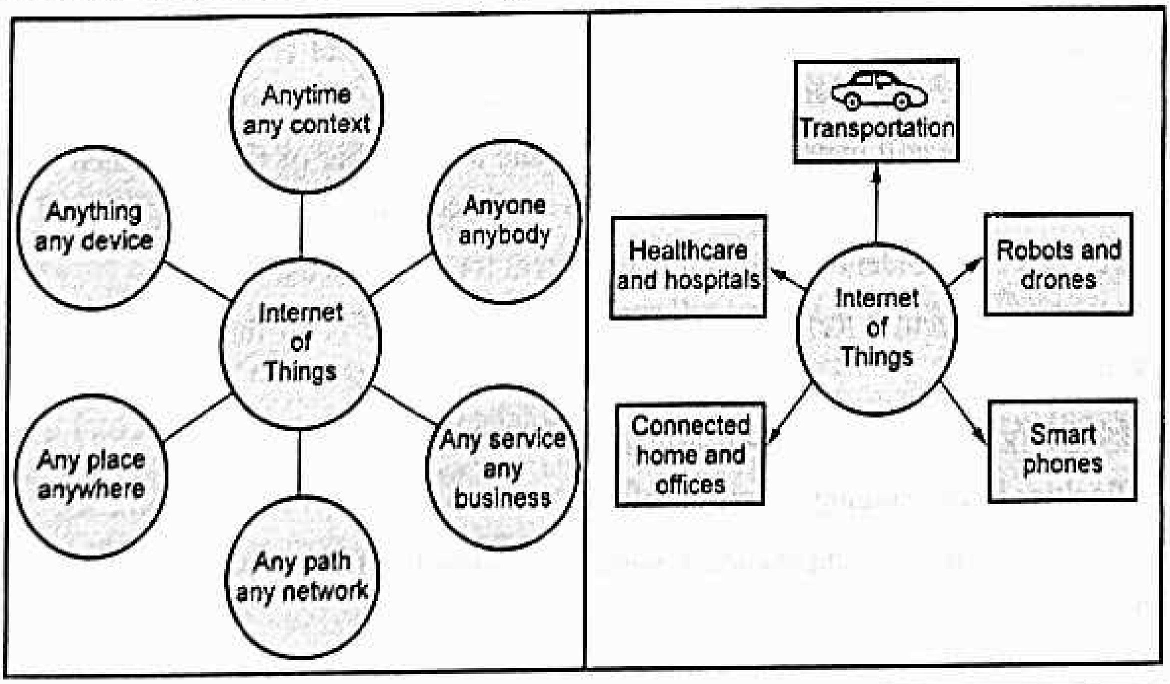


Fig. 2.1.1 : Evolutionary phase of internet

- Evolutionary phase of internet is Connectivity, Networked Economy, Immersive Experiences and IoT.
  1. Connectivity : In the phase, peoples are connected to email, web services and searches the information.
  2. Networked economy : This phase support e-commerce and supply chain enhancements along with collaborative engagement to drive increased efficiency in business processes.
  3. Immersive experiences : This phase extended the Internet experience to encompass widespread video and social media while always being connected through mobility.
  4. Internet of things : It adds connectivity to objects and machines in the world around us to enable new services and experiences.

2.1.1 Definition of IoT

- The Internet of Things (IoT) is the network of physical objects i.e. devices, vehicles, buildings and other items embedded with electronics, software, sensors, and network connectivity that enables these objects to collect and exchange data.
- Wikipedia definition : The internet of things, also called the internet of objects, refers to a wireless network between objects, usually the network will be wireless and self-configuring, such as household appliances.
- WSIS 2005 definition : By embedding short-range mobile transceivers into a wide array of additional gadgets and everyday items, enabling new forms of communication between people and things, and between things.
- The internet of things refers to the capability of everyday devices to connect to other devices and people through the existing Internet infrastructure.



- Devices connect and communicate in many ways. Examples of this are smart phones that interact with other smart phones, vehicle-to-vehicle communication, connected video cameras, and connected medical devices. They are able to communicate with consumers, collect and transmit data to companies, and compile large amounts of data for third parties.
- IoT data differs from traditional computing. The data can be small in size and frequent in transmission. The number of devices, or nodes, that are connecting to the network are also greater in IoT than in traditional PC computing.
- Machine-to-Machine communications and intelligence drawn from the devices and the network will allow businesses to automate certain basic tasks without depending on central or cloud based applications and services.
- IoT impacts every business. Mobile and the Internet of Things will change the types of devices that connect into a company's systems. These newly connected devices will produce new types of data.
- The Internet of Things will help a business gain efficiency, harness intelligence from a wide range of equipment, improve operations and increase customer satisfaction.
- Ubiquitous computing, pervasive computing, Internet Protocol, sensing technologies, communication technologies, and embedded devices are merged together in order to form a system where the real and digital worlds meet and are continuously in symbiotic interaction.
- The smart object is the building block of the IoT vision. By putting intelligence into everyday objects, they are turned into smart objects able not only to collect information from the environment and interact /control the physical world, but also to be interconnected, to each other, through Internet to exchange data and information.
- The expected huge number of interconnected devices and the significant amount of available data open new opportunities to create services that will bring tangible benefits to the society, environment, economy and individual citizens.
- However, the IoT is still maturing, in particular due to a number of factors, which limit the full exploitation of the IoT. Some of the factors are listed below :
  1. There is no unique identification number system for object in the world.
  2. IoT uses Architecture Reference Model (ARM) but there is no further development in ARM.
  3. Missing large-scale testing and learning environments.
  4. Difficulties in exchanging of sensor information in heterogeneous environments.
  5. Difficulties in developing business which embraces the full support of the Internet of Things.

2.1.2 IoT Characteristics

1. Interconnectivity : Everything can be connected to the global information and communication infrastructure.
2. Heterogeneity : Devices within IoT have different hardware and use different networks but they can still interact with other devices through different networks.
3. Things-related services : Provides things-related services within the constraints of things, such as privacy and semantic consistency between physical and virtual thing.
4. Dynamic changes : The state of a device can change dynamically, thus the number of devices can vary.
5. Integrated into information network : IoT devices are integrated with information network for communication purpose. It will exchange data with other devices.
6. Self-adapting: Self-Adaptive is a system that can automatically modify itself in the face of a changing context, to best answer a set of requirements.
7. Self-configuration primarily consists of the actions of neighbour and service discovery, network organization, and resource provisioning.

2.1.3 Component of IoT

- The hardware utilized in IoT systems includes devices for a remote dashboard, devices for control, servers, a routing or bridge device, and sensors. These devices manage key tasks and functions such as system activation, action specifications, security, communication, and detection to support-specific goals and actions.
- Major components of IoT devices are as follows :

1. **Control units :** A small computer on a single integrated circuit containing processor core, memory and a programmable I/O peripheral. It is responsible for the main operation.
2. **Sensor :** Devices that can measure a physical quantity and convert it into a signal, which can be read and interpreted by the microcontroller unit. These devices consist of energy modules, power management modules, RF modules, and sensing modules. Most sensors fall into 2 categories : Digital or analog. An analog data is converted to digital value that can be transmitted to the Internet.

   a. Temperature sensors : accelerometers

   b. Image sensors : gyroscopes

c. Light sensors : acoustic sensors

d. Micro flow sensors : humidity sensors

e. Gas RFID sensors : pressure sensors

3. **Communication modules :** These are the part of devices and responsible for communication with rest of IoT platform. They provide connectivity according to wireless or wired communication protocol they are designed. The communication between IoT devices and the Internet is performed in two ways :

   a) There is an Internet-enable intermediate node acting as a gateway;

   b) The IoT Device has direct communication with the Internet.

   - The communication between the main control unit and the communication module uses serial protocol in most cases.

4. **Power Sources :** In small devices the current is usually produced by sources like batteries, thermocouples and solar cells. Mobile devices are mostly powered by lightweight batteries that can be recharged for longer life duration.

   - **Communication Technology and Protocol :** IoT primarily exploits standard protocols and networking technologies. However, the major enabling technologies and protocols of IoT are RFID, NFC, low-energy Bluetooth, low-energy wireless, low-energy radio protocols, LTE-A, and WiFi-Direct. These technologies support the specific networking functionality needed in an IoT system in contrast to a standard uniform network of common systems.

2.1.4 Working of IoT

- Fig. 2.1.2 shows working of IoT.

1. **Collect and transmit data :** The device can sense the environment and collect information related to it and transmit it to a different device or to the Internet.

2. **Actuate device based on triggers :** It can be programmed to actuate other devices based on conditions set by user.

3. **Receive information :** Device can also receive information from the network.

4. **Communication assistance :** It provides communication between two devices of same network or different network.

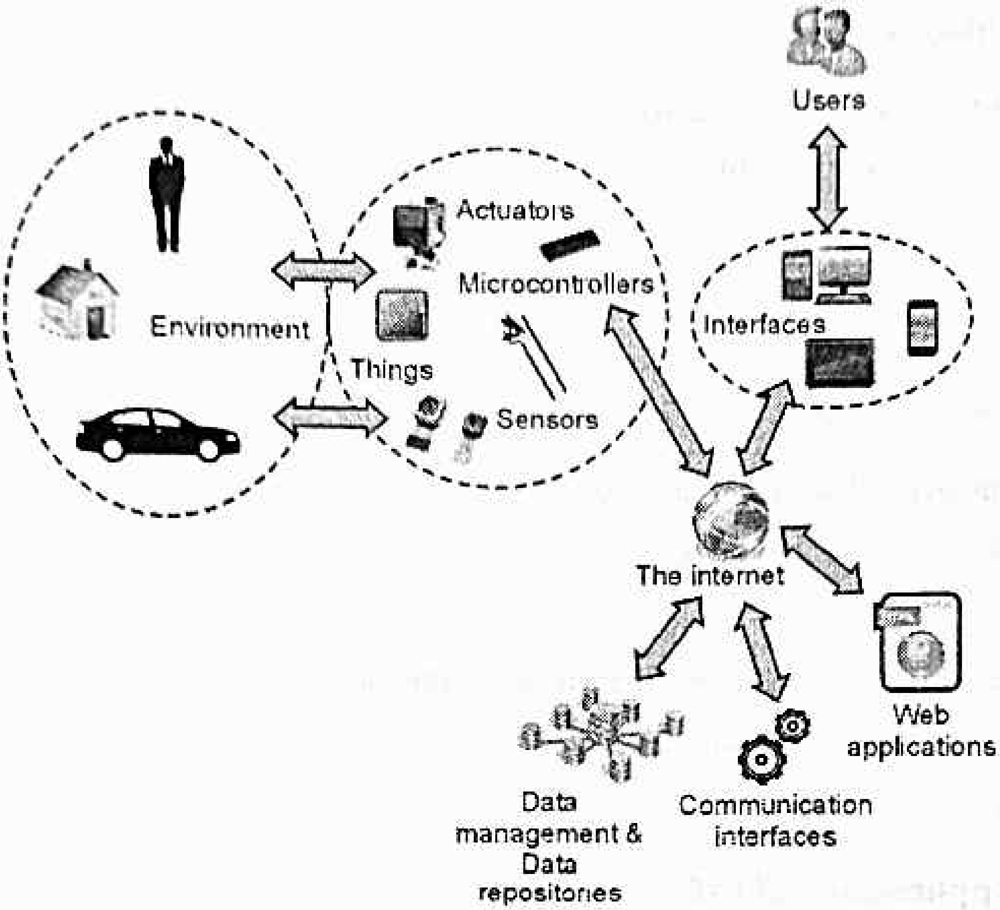


Fig. 2.1.2 : Working of IoT

- Sensors for various applications are used in different IoT devices as per different applications such as temperature, power, humidity, proximity, force etc.
- Gateway takes care of various wireless standard interfaces and hence one gateway can handle multiple technologies and multiple sensors.
- The typical wireless technologies used widely are 6LoWPAN, Zigbee, Zwave, RFID, NFC etc. Gateway interfaces with cloud using backbone wireless or wired technologies such as WiFi, Mobile , DSL or Fibre.

2.1.5 Advantages and Disadvantages

Advantages of IoT

1. Improved customer engagement and communication.
2. Support for technology optimization
3. Support wide range of data collection
4. Reduced waste

Disadvantages of IoT

1. **Loss of privacy and security :** As all the household appliances, industrial machinery, public sector services like water supply and transport, and many other devices all are connected to the Internet, a lot of information is available on it. This information is prone to attack by hackers.

2. **Flexibility :** Many are concerned about the flexibility of an IoT system to integrate easily with another.

3. **Complexity :** The IoT is a diverse and complex network. Any failure or bugs in the software or hardware will have serious consequences. Even power failure can cause a lot of inconvenience.

4. **Compatibility :** Currently, there is no international standard of compatibility for the tagging and monitoring equipment.

5. Save time and money.

2.1.6 Applications of IoT

1. **Home :** Buildings where people live. It controls home and security systems.

2. **Offices :** Energy management and security in office buildings; improved productivity, including for mobile employees.

3. **Factories :** Places with repetitive work routines, including hospitals and farms; operating efficiencies, optimizing equipment use and inventory.

4. **Vehicles :** Vehicles including cars, trucks, ships, aircraft, and trains; condition-based maintenance, usage-based design, pre-sales analytics

5. **Cities :** Public spaces and infrastructure in urban settings; adaptive traffic control, smart meters, environmental monitoring, resource management.

6. **Worksites :** It is custom production environments like mining, oil and gas, construction; operating efficiencies, predictive maintenance, health and safety.

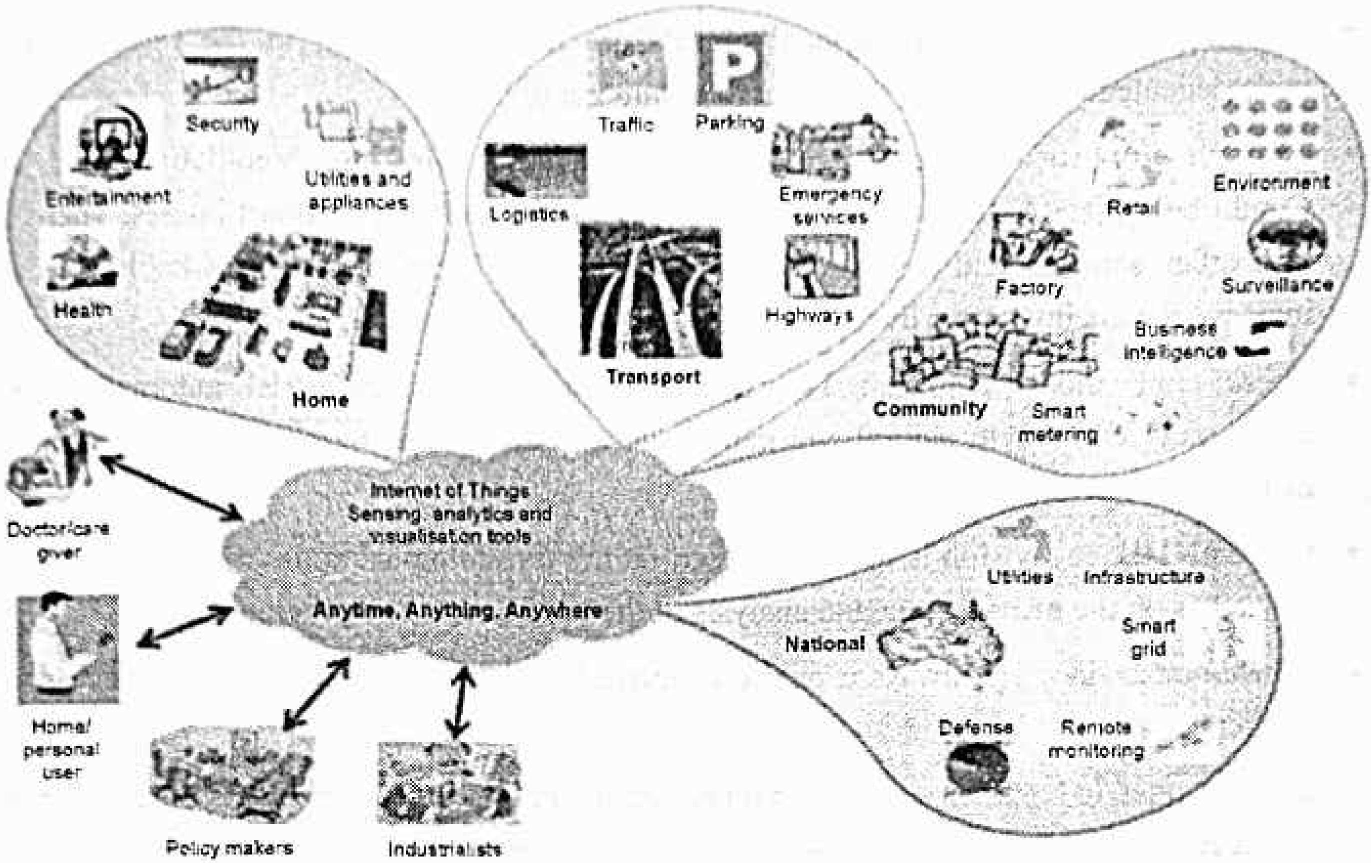


Fig. 2.1.3

Review Questions

1. *Why do IoT systems have to be self-adapting and self-configuring ?* **SPPU : May-18, Marks 2**
2. *Draw and explain block diagram of IoT device.* **SPPU : May-18, Marks 4**
3. *Define internet of things and briefly explain its characteristics.* **SPPU : Dec.-19, Marks 6**
4. *Define and explain horizontal, vertical application and 5A and 3I.* **SPPU : March-20, Marks 5**

2.2 Trends in Adoption of IoT **SPPU : Dec.-18**

- With the advent of the Internet of Things era, homes, cities and almost everything is becoming smart. Those smart objects can sense the physical world by obtaining data from sensors, affecting the sensed world by triggering actions using actuators, engage users by interacting with them whenever necessary, and process gathered data to provide useful information to us.

- An increasing number of physical objects are being connected to the Internet at an unprecedented rate realizing the idea of the Internet of Things.
- Examples of such objects include thermostats and Heating, Ventilation, and Air Conditioning (HVAC) monitoring and control systems that enable smart homes. There are also other domains and environments in which the IoT can play a remarkable role and improve the quality of our lives.
- These applications include : Transportation, health care, industrial automation, and emergency response to natural and man-made disasters where human decision making is difficult.
- Currently, the IoT vision is mainly focused on the technological and infrastructure aspect, as well as on the management and analysis of the huge amount of generated data.
- Consumer devices are always a public concern, but there are currently five types of IoT applications :
  a) Consumer IoT : Such as light fixtures, home appliances, and voice assistance for the elderly.
  b) Commercial IoT : IoT applications in the healthcare and transport industries, such as smart pacemakers, monitoring systems, and vehicle to vehicle communication (V2V).
  c) The Industrial Internet of Things : IIoT includes digital control systems, statistical evaluation, smart agriculture, and big industrial data.
  d) Infrastructure IoT enables the connectivity of smart cities through infrastructure sensors, management systems, and user-friendly user apps.
  e) Military Things (IoMT) applying IoT technologies in the military field, such as robots for surveillance and human-wearable biometrics for combat.

2.2.1 Consumer IoT

- Consumer IoT refers to the billions of physical personal devices, such as smartphones, wearables, fashion items and the growing number of smart home appliances, that are now. Connected to the internet, collecting and sharing data.
- Consumers want IoT products that fit seamlessly into their daily lives and simplify routine tasks. Finding keys, locking the front door, turning lights on and off, these tasks can be controlled with wireless technologies.

- Fig. 2.2.1 shows consumer IoT.

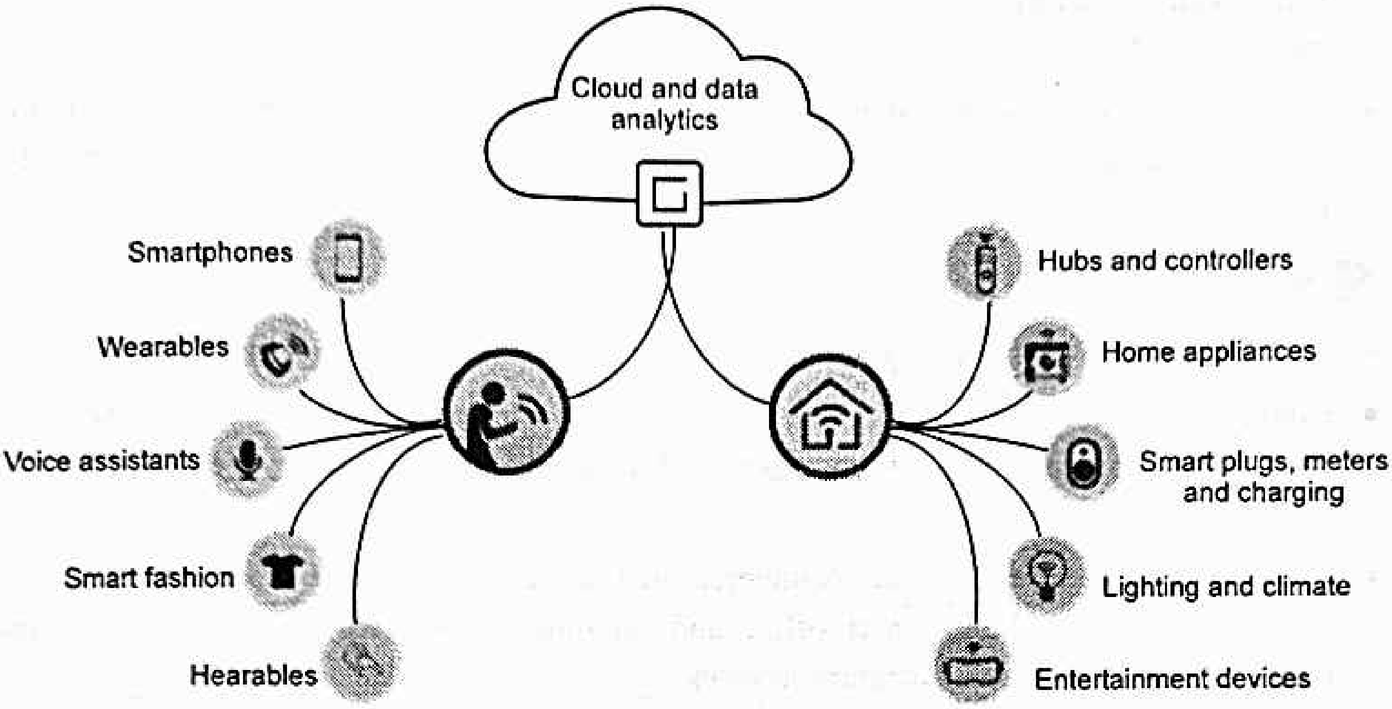


Fig. 2.2.1 : Consumer IoT

- Consumer smart assistants, such as Amazon Alexa, Google Home and Apple Homepod, are the hubs for consumer IoT. They build on stable technologies : Bluetooth (BLE,) Wi-Fi and derivatives of ARM mobile processors. On the software side, they all rely on Natural Language Processing (NLP) and acoustics that are fairly mature.
- The real-time data collected from connected devices and smart connected consumer products can generate insights and provide personalized information and services to the consumers.
- Smart homes, connected vehicles, wellness and fitness through wearables, smart retail and smart farming are a few examples of how businesses can achieve their short - term and long - term goals.
- Business-to-consumer value : Using product data to inform better customer experience, personalization, reliability, preemptive support, convenience, security, product/service improvement, etc.
- Business-to-business value : Using product data to inform operational efficiencies, improve sales conversion, marketing decisions, security, inventory optimization, etc.
- Business-to-ecosystem value : Using product data to inform partner strategies, integrations, APIs, content needs, supply chain efficiencies, etc.

- Consumer IoT has long held a reputation for poor security standards. Homes today are filled with connected devices. It's not just smart speakers, it's app - enabled espresso machines and wifi - connected security cameras.
- The majority of IoT devices purchased for the home are relatively cheap and little effort is made to protect them at a hardware or software level at this end of the spectrum by manufacturers.

2.2.2 Commercial IoT

- Commercial IoT lies between consumer IoT and industrial IoT applications.
- Commercial IoT refers to devices and systems for business and enterprise use. These can vary broadly by sector. IoT is increasingly leveraged in healthcare, event venues, office buildings, and more.
- Specifically, when we talk about commercial IoT some of the key use cases that we see are intelligent asset tracking, smart office and buildings, connected lighting, sensing and monitoring of all types and location services.

2.2.3 Industrial IoT

- Industrial IoT (IIoT) infrastructure should be protected by a comprehensive security solution (device-to-cloud) that does not disrupt operations, service reliability or profitability.
- Industrial IoT refers to devices and systems used in manufacturing or industrial purposes. Examples include supply chain asset tracking and fleet tracking.
- Fig. 2.2.2 shows industrial IoT system architecture. (Refer Fig. 2.2.2 on next page)
- The common requirements of the IoT are technical requirements independent of any specific application domain.
- IoT non-functional requirements refer to the requirements related to the implementation and operation of the IoT itself , i.e. Interoperability, scalability, reliability, high availability, adaptability, manageability.
- IoT functional requirements refer to the requirements related to the IoT actors, these requirements have been categorized as,

  1. Application support requirements
  2. Service requirements
  3. Communication requirements
  4. Device requirements

5. Data management requirements
6. Security and privacy protection requirements.

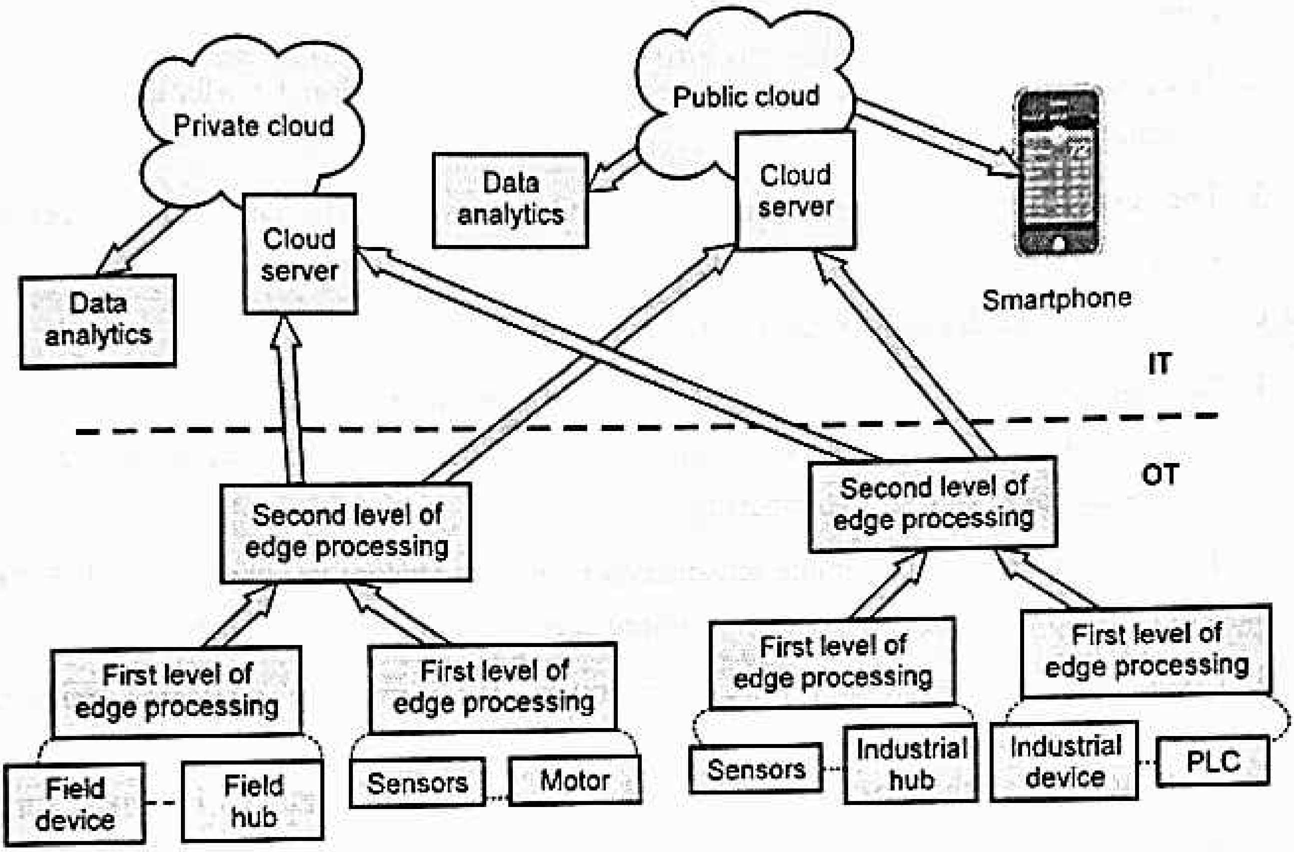


Fig. 2.2.2 : Industrial IoT system architecture

- The IoT application capabilities for industrial application should meet requirements such as :
  1. Reliability : Reliable IoT devices and systems should allow a continuous operation of industrial processes and perform on-site activities
  2. Robustness : The IoT application and devices should be robust and adapted to the task and hard working conditions
  3. Simple to use
  4. Low maintains cost
  5. Support optimal and adaptive set of features
  6. It must have reach sensing and data capabilities
  7. Industry grade support and services. The IoT applications should be supported over years in operation by a set of rich tools and continuously updated services
  8. Support standardization
  9. Security and safety.

- The industrial internet of things will be shaped by the appearance of three features :
  1. The enhancement of basic mechanical devices through sensors and other data producing devices ("smartness");
  2. The possibilities of ever faster and more flexible data allocation, transfer and processing (computing capacities);
  3. The ever increasing digital interconnectivity between the engaged devices and computational capacities (digital integration).

❑ Challenges faced by IoT industry applications

1. Security. As the IoT connects more devices together, it provides more decentralized entry points for malware. Less expensive devices that are in physically compromised locales are more subject to tampering.
2. Trust and Privacy. With remote sensors and monitoring a core use case for the IoT, there will be heightened sensitivity to controlling access and ownership of data.
3. Complexity, confusion and integration issues. With multiple platforms, numerous protocols and large numbers of APIs, IoT systems integration and testing will be a challenge to say the least. The confusion around evolving standards is almost sure to slow adoption.
4. Evolving architectures competing standards. With so many players involved with the IoT, there are bound to be ongoing turf wars as legacy companies seek to protect their proprietary systems advantages and open systems proponents try to set new standards. There may be multiple standards that evolve based on different requirements determined by device class, power requirements, capabilities and uses. This presents opportunities for platform vendors and open source advocates to contribute and influence future standards.
5. Concrete use cases and compelling value propositions. Lack of clear use cases or strong ROI examples will slow down adoption of the IoT. Although technical specifications, theoretical uses and future concepts may suffice for some early adopters, mainstream adoption of IoT will require well-grounded, customer-oriented communications and messaging around "what's in it for me."
6. There are several wireless standards which can be used to connect devices to a network, most are still developing. That means delays as products play catch - up with new networking standards.

2.2.4 Infrastructure IoT

- Five things are making up an IoT infrastructure are IoT platforms, access technologies, data storage and processing, data analytics, and security. An essential part of any internet - connected device, these five pillars are what enable growth for future IoT solutions.

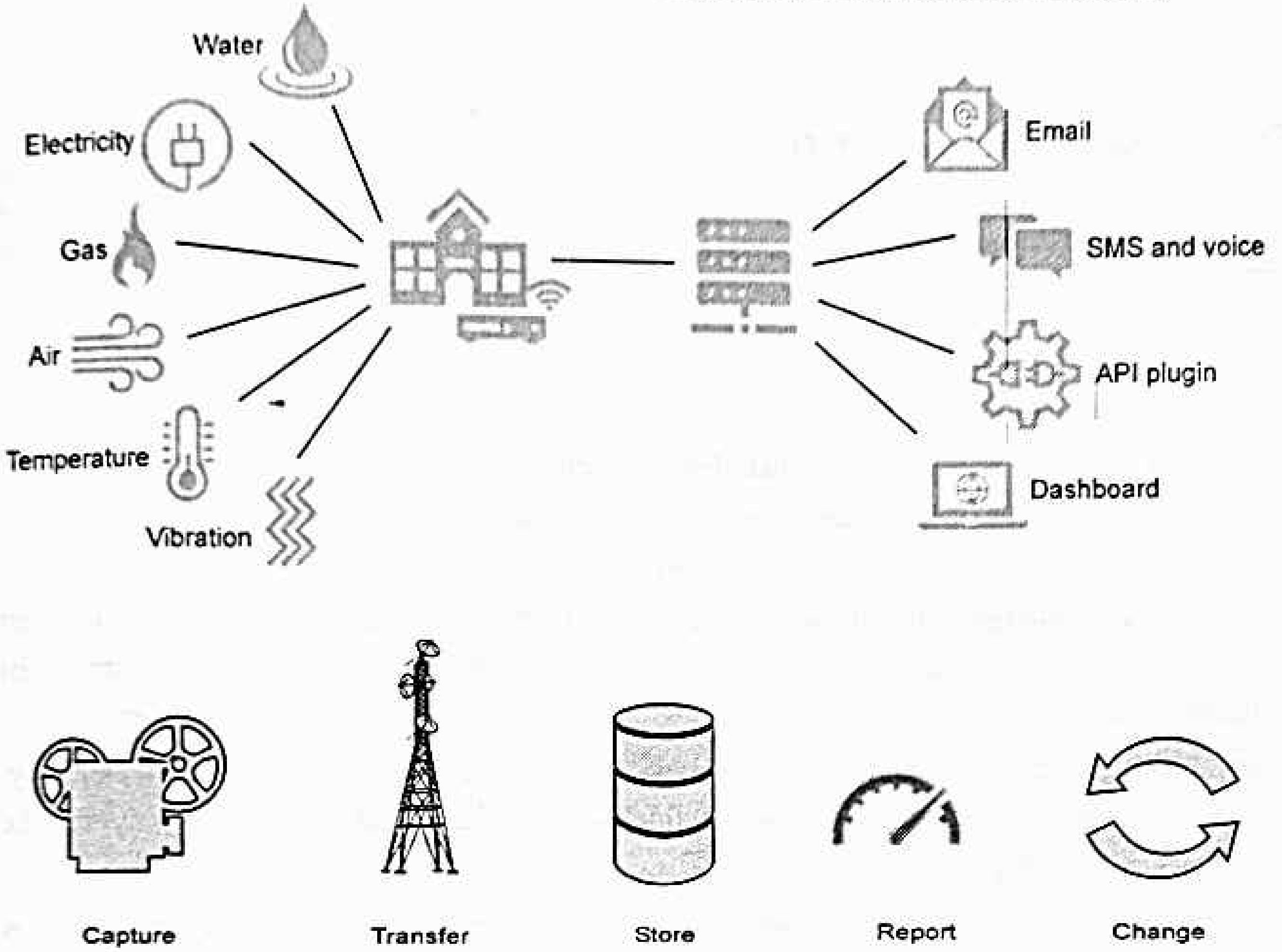


Fig. 2.2.3

- Smart infrastructure also includes implementing charging stations in parking systems, city fleets, shopping malls and buildings, airports, and bus stations across the city. Electronic Vehicle (EV) charging platforms can be integrated with IoT to streamline the operations of EV charging and addresses the impact of the power grid.
- A simple IoT infrastructure can be broken down into five key areas, capture, transfer, store, report and change.
- Capture covers the type of data source and the way the information is captured. This could be temperature or power consumption using digital, analogue or Modbus inputs. This data can then be transferred directly to the cloud using LoRaWAN for example or it can be transmitted to a gateway first using RF 868 mHz and then from the gateway to the cloud using GPRS 2G/3G.

- The data could be stored on the IoT supplier's cloud or sent via an API to the user's own cloud. From the cloud the data can be Reported and ingested in several ways. This could be in the form of a dashboard, an email alert or via an API into existing reporting tools.
- Once the information has been ingested it can be used to influence some form of change may that be a physical input or an automated response back via the IoT infrastructure or another system.

2.2.5 Military Things (IoMT)

- The IoMT is the application of Internet of Things (IoT) technologies and concepts to the military domain. Military IoT adoption is still in its infancy, however defence companies and the armed forces are eager to prepare, understand and leverage the IoT.
- There are clear military benefits from the use of IoT devices for the armed forces, ranging from vehicle maintenance to personnel monitoring to stock control.
- In today's scenarios military commanders may have only several minutes to be presented with an assessment of the situation in a way that is quickly assimilated, to assess potential courses of action and to make their decision.
- It needs to draw upon all possible sources to ensure that the most complete and relevant picture can be created of the situation, and the implications of different decisions understood.
- However, the integration of heterogeneous sensors and systems diverse in technology, environmental constraints and level of fidelity is a challenging issue not only for the military organizations.
- To provide a response to these challenges, the concept of Internet of Things (IoT) is extensively developed world-wide with focus on civilian applications. IoT is considered as an integral part of Future Internet and could be defined as a **dynamic global network infrastructure** with self-configuring capabilities based on standard and interoperable communications protocols where physical and virtual things have identities, physical attributes and virtual personalities and use intelligent interfaces, and are seamlessly integrated into the information network.
- The concept of IoT developed for civilian market does not fit the military requirements. Highly dynamic and disruptive environment, fragile nature of wireless communications, especially at lower tactical levels, specific requirements for speed and fidelity with which information must be exchanged and analysed requires robust and effective mechanisms, as well as specific applications that present IoT solutions do not provide.
- The Internet of military things refers to a network of sensors, wearables, and other devices that use cloud and edge computing to give advantage to a fighting force.

- Long-Term Evolution (LTE) is a standard for wireless broadband communication for mobile devices that increases the capacity and speed of data transfer and provides several add - on facilities.
- IoMT is a type of Internet of Things designed for combat operations and warfare based upon smart technology and artificial intelligence. It comprises a complex network of interconnected entities in the military domain that continually communicate with each other to accomplish a broad range of activities in a more efficient and informed manner.

Review Question

1. *List and explain different components of an IoT system.* **SPPU : Dec.-18, Marks 4**

2.3 IoT Device **SPPU : Dec.-18**

- IoT devices have unique identity and they are refer as "things" in IoT. Device can perform remote sensing, actuating and monitoring.
- IoT devices can exchange data between them and process data or send to centralized location for processing and storage. Fig. 2.3.1 shows block diagram of IoT device.

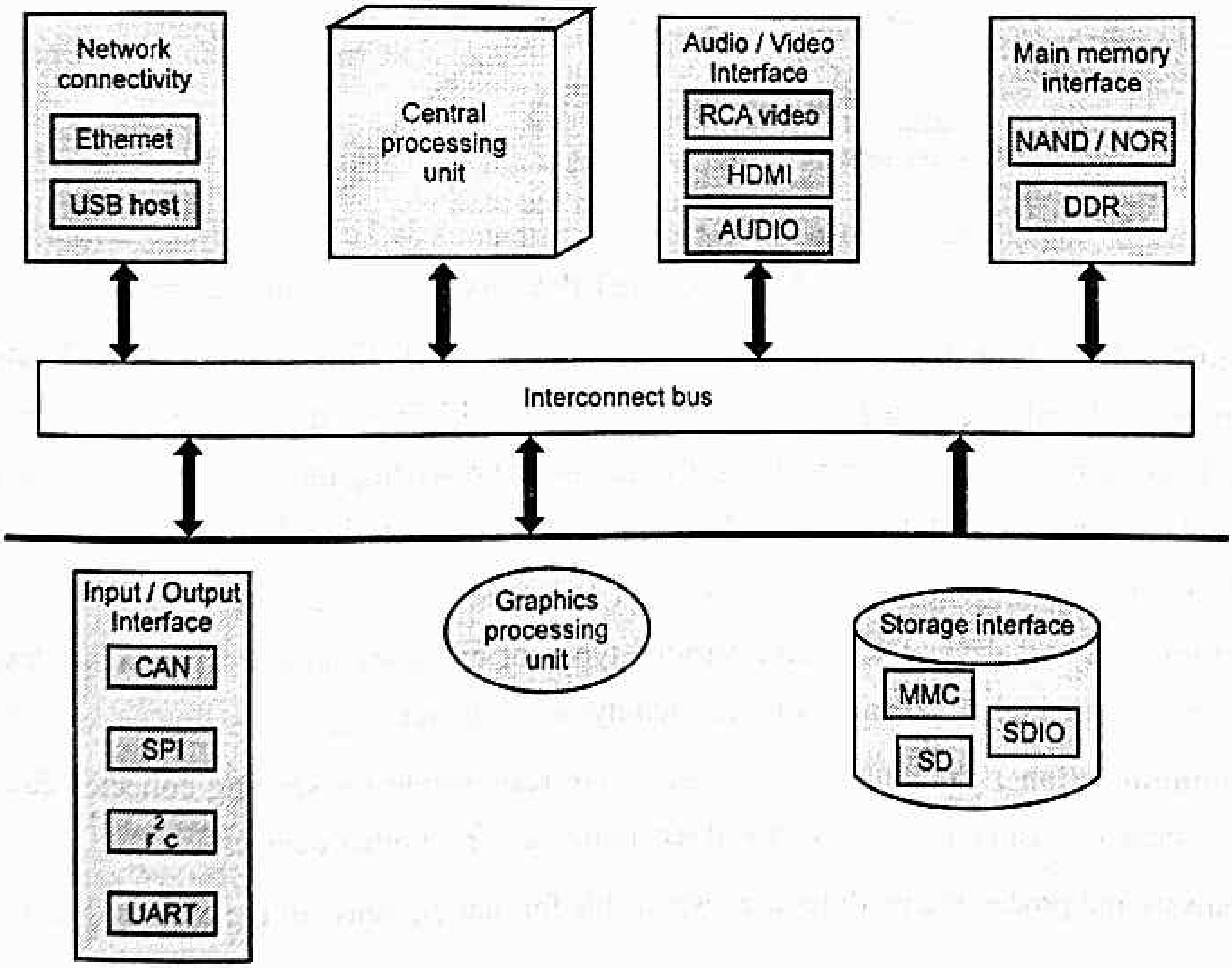


Fig. 2.3.1 : Block diagram of IoT device

- IoT devices provide interface to various wire and wireless devices. Interface includes memory interface, I/O interface for sensors, Internet connectivity interface, storage interface etc.
- Using sensors, IoT collects various information like temperature, light intensity, humidity, air pressure. Some application used cloud based storage. Collected information is stored in cloud and transmitted to other devices.
- Various types of IoT devices are smart clothing, smart watch, wearable sensors, LED lights, automobile industry etc. Fig. 2.3.2 shows IoT devices.

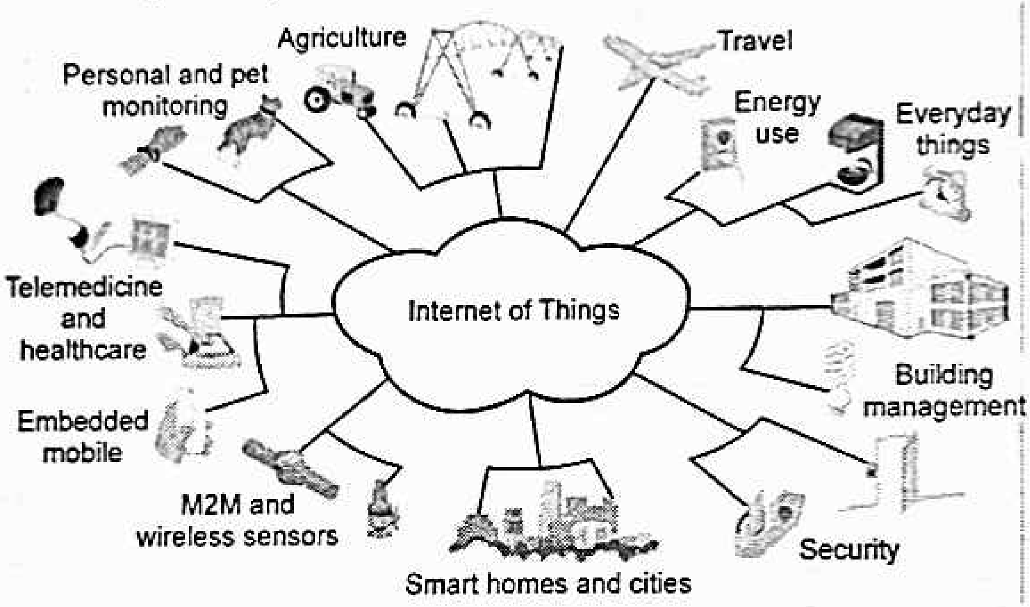


Fig. 2.3.2 : IoT devices

- **Sensor :** Devices that can measure a physical quantity and convert it into a signal, which can be read and interpreted by the microcontroller unit. These devices consist of energy modules, power management modules, RF modules and sensing modules. Most sensors fall into 2 categories : Digital or analog. An analog data is converted to digital value that can be transmitted to the Internet.
- **Actuation :** IoT devices can have various types of actuators attached that allow taking actions upon the physical entities in the vicinity of the device.
- **Communication :** Communication modules are responsible for sending collected data to other device or cloud based servers and receiving data from other devices.
- Analysis and processing modules are responsible for making sense of the collected data.

❑ IoT Device Life Cycle :

- IoT devices are generally more like single-purpose computers. The first life cycle, for example, includes four steps :

  1. **Boot-up :** The device loads the firmware and starts to work as defined.
  2. **Initialization :** Once boot-up is completed, the system reads the configuration, established connections, syncs up data, etc.
  3. **Operation :** The device performs its designed purpose continually.
  4. **Update :** New firmware is installed, the device reboots and then starts to load the new firmware.

- The device should complete its previous life cycle before starting the next life cycle every time the firmware is updated. Eventually, the device will be retired for whatever reason. When it does, it reaches the end of the device life cycle called termination.

Review Question

1. *Explain lifecycle of an IoT device.* **SPPU : Dec.-18, Marks 4**

2.4 IoT System Building Blocks

- The hardware utilized in IoT systems includes devices for a remote dashboard, devices for control, servers, a routing or bridge device, and sensors. These devices manage key tasks and functions such as system activation, action specifications, security, communication, and detection to support-specific goals and actions.

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❑ Working :

1. Collect and transmit data : The device can sense the environment and collect information related to it and transmit it to a different device or to the Internet.

2. Actuate device based on triggers : It can be programmed to actuate other devices based on conditions set by user.

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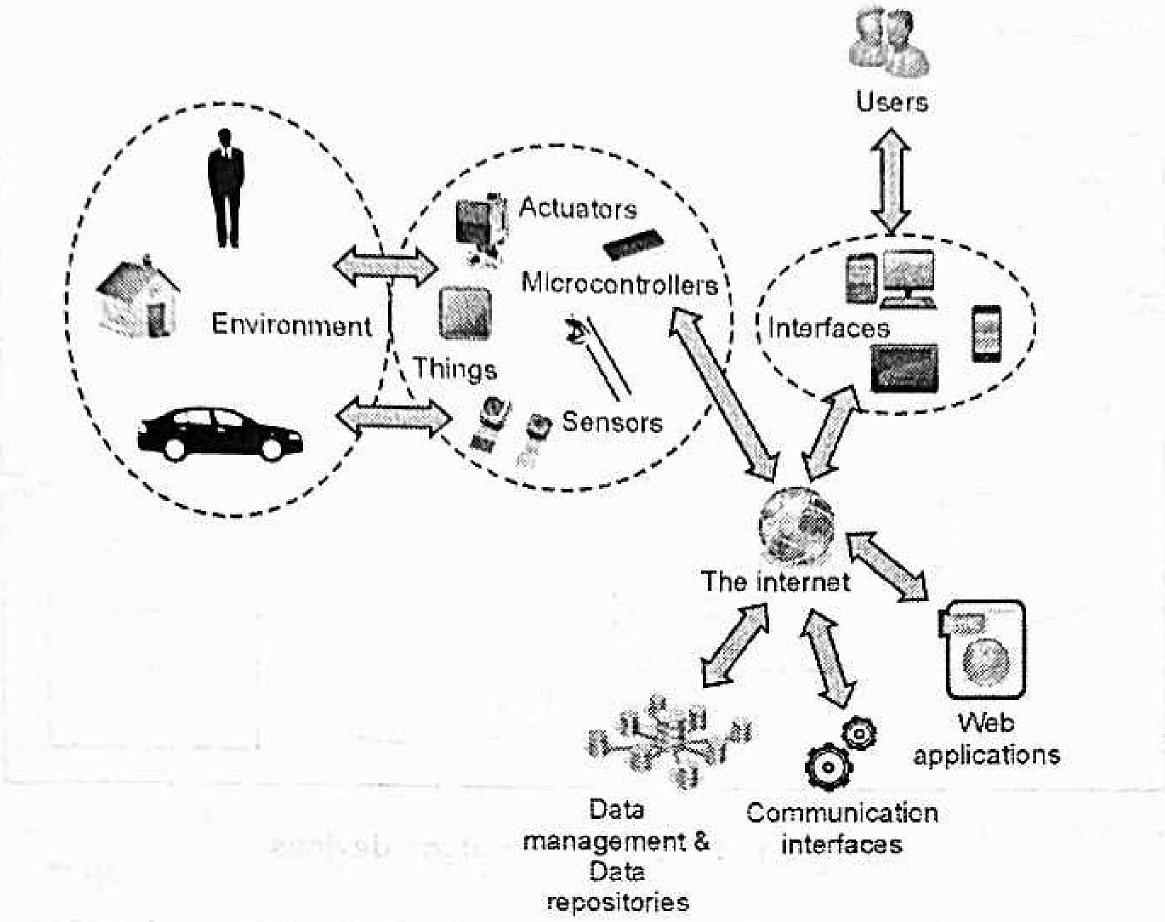
- Fig. 2.4.1 shows working of IoT.



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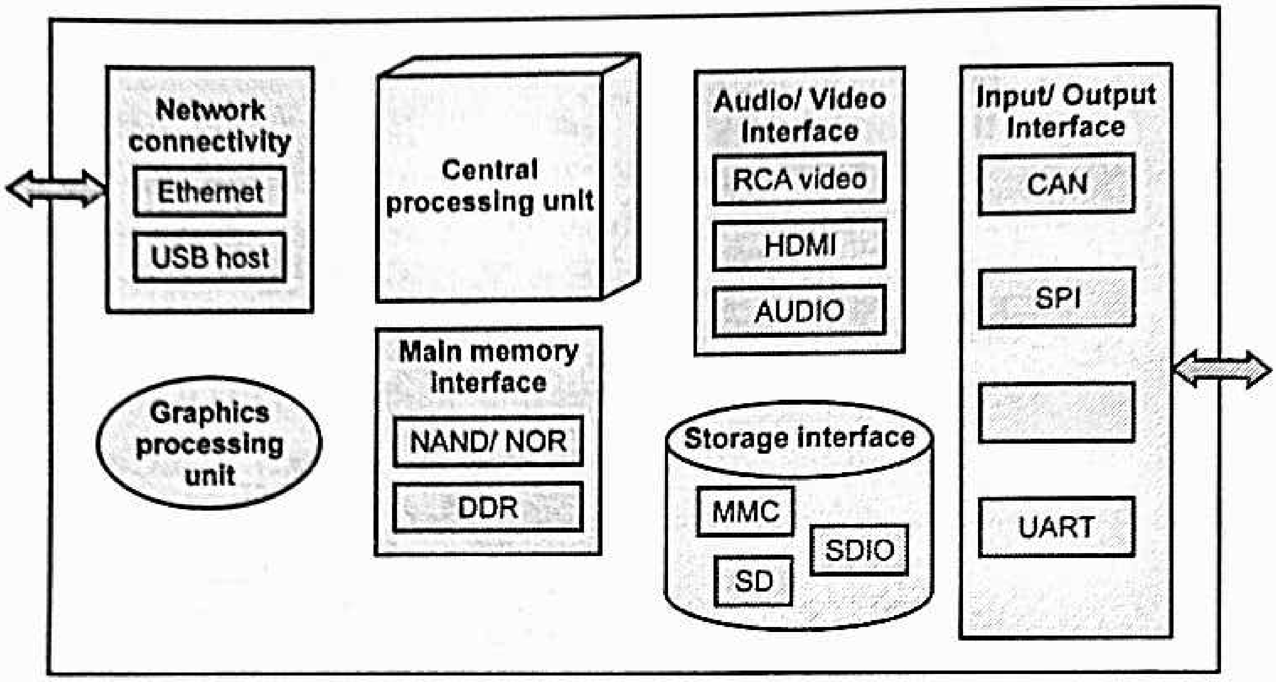
2.5 Physical Design of IoT

- Physical Design of IoT system refers to IoT Devices and IoT Protocols.

2.5.1 Things in IoT

- IoT devices have unique identity and they are refer as "things" in IoT. Device can perform remote sensing, actuating and monitoring. IoT devices can exchange data between them and process data or send to centralized location for processing and storage. Fig. 2.5.1 shows block diagram of IoT device.



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2.5.2 IoT Protocol

- Fig. 2.5.2 shows IoT protocols.

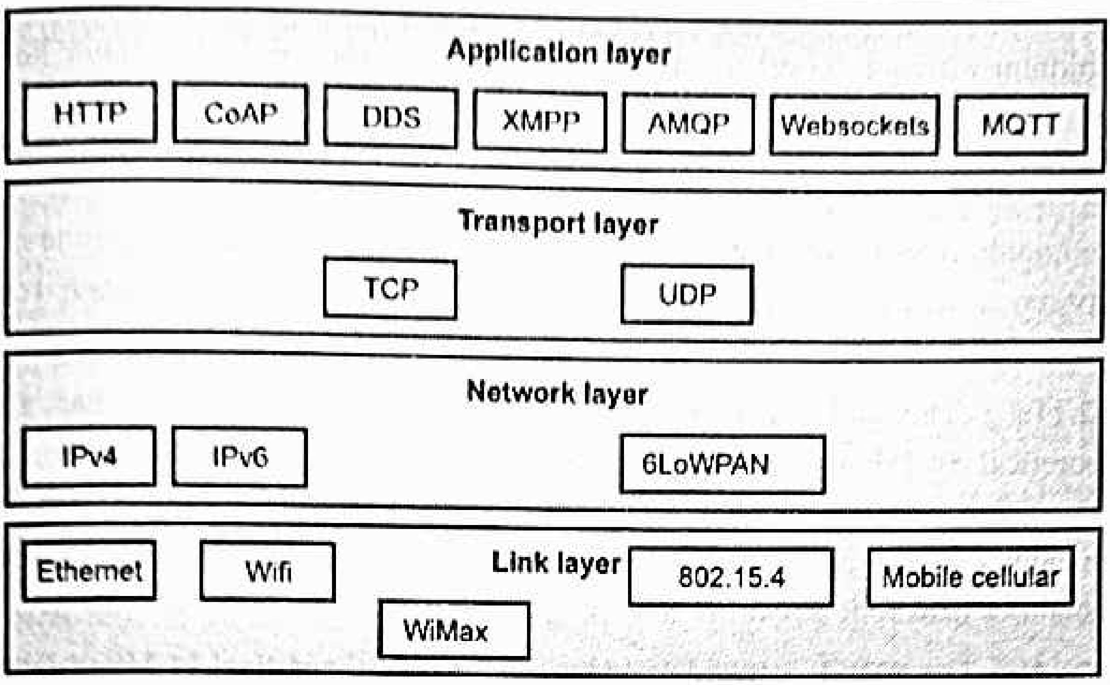

Fig. 2.5.2 : IoT protocol

1. Link Layer

- Link layer protocols decide how data is sent on physical medium. Link layer works within the local area network. Protocol of link layer is explained below :

a. 802.3 Ethernet

- This protocol is used for wired medium. Ethernet, in its most basic version runs at 10 Mbit/s. Ethernet has traditionally been used to network enterprise workstations and to transfer non-real-time data.
- The Ethernet standard allows for several different implementations such as twisted pair and coaxial cable. The maximum length of an Ethernet is determined by the node's ability to detect collisions.
- The worst case occurs when two nodes at opposite ends of the bus are transmitting simultaneously. Ethernet does not provide any mechanism for acknowledging received frames, making it what is known as an unreliable medium.
- Carrier sense multiple access with collision detection (CSMA/CD) is the most commonly used protocol for LANs. 10BASE5 is generally used as low cost alternative for fiber optic media for use as a backbone segment with in a single building.
- 10BASET is 10 MHz Ethernet running over UTP cable. It also uses passive star topology. The maximum cable segment allowed is 100 - 150 meters. There is no

minimum distance requirements between devices, such devices cannot be connected serially but in star wired

b. 802.11 Wifi

- Commonly referred to as Wi-Fi the 802.11 standards define a through-the-air interface between a wireless client and a base station access point or between two or more wireless clients.
- **802.11a :** The 802.11a standard uses the 5 GHz spectrum and has a maximum theoretical 54 Mbps data rate. The 5GHz spectrum has higher attenuation than lower frequencies, such as 2.4 GHz used in 802.11b/g standards. Products with 802.11a are typically found in larger corporate networks or with wireless Internet service providers in outdoor backbone networks.
- **802.11b :** The 802.11 standard provides a maximum theoretical 11 Mbps data rate in the 2.4 GHz Industrial, Scientific and Medical (ISM) band.
- 802.11b uses Complementary Code Keying (CCK) instead of Differential Quadrature Phase Shift Keying (DQPSK) used at lower rates.
- **802.11g :** It provides 20 Mbps and more in the 2.4 GHz band.

c. 802.16 WiMax

- WiMAX refers to broadband wireless networks that are based on the IEEE 802.16 standard, which ensures compatibility and interoperability between broadband wireless access equipment.
- The 802.16 astandard will support OFDM in the 2-to-11 GHz frequency range. The 802.16b standard will operate in the 5 GHz ISM band. A single WiMAX tower can provide coverage to a very large area as big as 3000 square miles
- WiMAX receiver : The receiver and antenna could be a small box or Personal Computer Memory card or they could be built into a laptop the way WiFi access is today.

d. 802.15.4 Zigbee

- In 2002, seeing that neither Wi-Fi nor Bluetooth could not fit some of their needs for embedded systems , a number of industrial companies formed the consortium called ZigBee Alliance, aimed at providing standards for low cost/low consumption wireless communications. Then, with the birth of IEEE 802.15.4 group.
- ZigBee communications can reach up to 500m, with a data rate of up to 250 kbs, for a typical power consumption of 125 to 400 μW.

- As ZigBee is based on IEEE 802.15.4, there is no wake-up signal, but slots for sleep or activity, or in asynchronous mode, devices sleeping anytime they have nothing to say, with an ever-vigilant coordinator.
- To use a ZigBee module with a microcontroller, you need to connect it to a UART. There are other, optional pins to use, including a number of analog inputs / digital IOs and a PWM output indicating the strength of the signal which you can directly connect to a LED pin for observation purposes.
- There are two modes of data transfer namely Beacon mode and Non Beacon mode.
- In Beacon mode, when the devices are not sending the data they may enter a low power state and reduces the power consumption.
- In Non-beacon mode, the end devices need to be wake up only while sending the data while the routers and coordinators need to be active most of the time.

e. Mobile Communication (2G/3G/4G)

- GSM frequencies originally designed on 900 MHz range, now also available on 800 MHz, 1800 MHz and 1900 MHz ranges. The backbone of a GSM network is a telephone network with additional cellular network capabilities
- 4G is also called as long term evolution. It's promises data transfer rates of 100 Mbps.

❑ 2. Network Layer

- The network layer is responsible for the delivery of packets from the source to destination.
- Network layer uses IP address to choose one host among millions of host. In network layer, datagram needs a destination IP address for delivery and a source IP address for a destination reply.

a. IPv4

- IP is used for communicating all Internet enabled devices. The transport layer is responsible for delivery of message from one process to another.
- The network does the host to destination delivery of individual packets considering it as independent packet. But transport layer ensures that the whole message arrives intact and in order with error control and process control.
- An IP address is a numeric identifier assigned to each machine on an IP network. IP address is a software address, not a hardware address, which is hard-coded in the machine or NIC.

- An IP address is made up of 32 bits of information. These bits are divided into four parts containing 8 bit each.
- IPv4 addresses are unique. Two devices on the internet can never have the same address at the same time.
- Packets in the IPv4 layer are called datagrams. A datagram is a variable length.

b. IPv6

- IPv6 addresses are 128 bits in length. Addresses are assigned to individual interface on nodes, not to the node themselves.
- A single interface may have multiple unique unicast addresses. The first field of any IPv6 address is the variable length format prefix, which identifies various categories of addresses.

c. 6LoWPAN

- IPv6 over Low power Wireless Personal Area Network enables IPv6 in low-power and lossy wireless networks such as WSNs.
- 6LoWPAN defines header compression mechanisms.

❑ 3. Transport Layer

- A transport layer protocol provides for logical communication between application processes running on different hosts.
- The transport layer is responsible for delivery of message from one process to another. The network does the host to destination delivery of individual packets considering it as independent packet.
- But transport layer ensures that the whole message arrives intact and in order with error control and process control.
- A transport protocol can offer reliable data transfer service to an application even when the underlying network protocol is unreliable, even when the network protocol loses, garbles and duplicate packets.

a. TCP (Transmission Control Protocol)

- TCP is the connection oriented protocol whereas UDP is connectionless protocol. Both are internet protocols used in the transport layer.
- TCP provides a connection-oriented, reliable, byte stream service. The term connection oriented means the two applications using TCP must establish a TCP connection with each other before they can exchange data.

- TCP does not support multicasting and broadcasting. The application data is broken into what TCP considers the best sized chunks to send. The unit of information passed by TCP to IP is called a segment.
- When TCP sends a segment it maintains a timer, waiting for the other end to acknowledge reception of segment. If an acknowledgement isn't received in time, the segment is retransmitted.

b. (UDP) User Datagram Protocol

- UDP is a simple, datagram-oriented, transport layer protocol. This protocol is used in place of TCP.
- UDP is connectionless protocol provides no reliability or flow control mechanisms. It also has no error recovery procedures.
- UDP makes use of the port concept to direct the datagrams to the proper upper-layer applications. UDP serves as a simple application interface to the IP.
- UDP uses port numbers as the addressing mechanism in the transport layer.

❑ 4. Application Layer

- Application layer is responsible for accessing the network by user. It provides user interfaces and other supporting services such as e-mail, remote file access, file transfer, sharing database, message handling (X.400), directory services (X.500).

a. HTTP (HyperText Transport Protocol)

- HTTP is an application protocol. HTTP is used to retrieve Web pages from remote servers.
- HTTP uses the services of TCP. HTTP is a stateless protocol. The client initializes the transaction by sending a request message. The server replies by sending a response.
- HTTP includes commands such as GET, PUT, POST, HEAD, DELETE, MOVE, LINK and UNLINK.
- HTTP messages are two types : Request and Response
- URL is a standard for specifying any kind of information on the internet. HTTP uses URL.

b. CoAP - Constrained Application Protocol

- CoAP is a specialized web transfer protocol for use with constrained nodes and constrained (e.g., low-power, lossy) networks.

- CoAP is designed for simplicity, low overhead and multicast support in resource-constrained environments.
- CoAP is a web protocol that runs over the UDP for IoT. Datagram Transport Layer Security (DTLS) is used to protect CoAP transmission.
- The protocol is designed for machine-to-machine (M2M) applications such as smart energy and building automation.
- CoAP provides a request/response interaction model between application endpoints, supports built-in discovery of services and resources, and includes key concepts of the Web such as URIs and Internet media types.
- CoAP is designed to easily interface with HTTP for integration with the Web while meeting specialized requirements such as multicast support, very low overhead, and simplicity for constrained environments.

c. Websocket

- WebSocket is a communications protocol, providing full-duplex communication channels over a single TCP connection. The WebSockets protocol does not run over HTTP, instead it is a separate implementation on top of TCP.
- The WebSocket protocol starts by a handshake in order to start the communication and exchange messages formatted as frames.
- After connection is established, messages can be transmitted, either client or server initiated. This means that you can make a dynamic web page where changes occur in real time.
- In that way Websockest communication presents a suitable protocol for IoT world where changes are usually asynchronously occurring and number of clients can be very large.

d. MQTT (Message Queue Telemetry Transport)

- MQTT is Open Connectivity for Mobile, M2M and IoT. MQTT is designed for high latency, low-bandwidth or unreliable networks. The design principle minimizes the network bandwidth and device resource requirements.
- MQTT is a lightweight broker-based publish/subscribe messaging protocol designed to be open, simple, lightweight and easy to implement.
- The MQTT protocol works by exchanging a series of MQTT control packets in a defined way. Each control packet has a specific purpose and every bit in the packet is carefully crafted to reduce the data transmitted over the network.

- A MQTT topology has a MQTT server and a MQTT client. MQTT control packet headers are kept as small as possible.
- Having a small header overhead makes this protocol appropriate for IoT by lowering the amount of data transmitted over constrained networks.
- MQTT is the protocol built for M2M and IoT which is used to provide new and revolutionary performance.

e. XMPP (Extensible Messaging Presence Protocol)

- The XMPP is targeted at delivering instant messages and presence information. It is an open and XML - Based protocol.
- Instant messaging (IM) is a service, where communicating parties typically end users send messages in one- to-one or one -to -many fashion in near real - time.
- An open technology for real-time communication, which powers a wide range of applications including instant messaging, presence, multi-party chat, voice and video calls, collaboration, lightweight middleware, content syndication, and generalized routing of XML data.
- XMPP support server-to-server communication and client-to-server communication.

f. DDS (Data Distribution Service)

- The first open international middleware standard directly addressing publish-subscribe communications for real-time and embedded systems.
- The DDS is an Object Management Group (OMG) standard for Pub/Sub that addresses the needs of mission and business critical applications, such as, financial trading, air traffic control and management, and complex supervisory and telemetry systems.
- DDS provides a shared "global data space" where any application can publish the data it has & subscribe to the data it needs.
- DDS is highly configurable by means of QoS settings. Heterogeneous systems can be easily accommodated.

g. AMQP (Advanced Message Queuing Protocol)

- A protocol to communicate between clients and messaging middleware servers (brokers). The Broker is the AMQP Server.
- AMQP supports both publish-subscribe model and point-to-point communication, routing and queuing.

- AMQP divides the brokering task between exchanges and message queues, where the first is a router that accepts incoming messages and decides which queues to route the messages to, and the message queue stores messages and sends them to message consumers.
- AMQP supports username and password authentication as well as SASL authorization. It also supports TLS encryption.

2.6 Sensors

- Sensor converts a physical quantity into a corresponding voltage. Sensor is a device that when exposed to a physical phenomenon (temperature, displacement, force, etc.) produces a proportional output signal (electrical, mechanical, magnetic, etc.).
- The term transducer is often used synonymously with sensors. Sensor is a device that responds to a change in the physical phenomenon. On the other hand, a transducer is a device that converts one form of energy into another form of energy. Sensors are transducers when they sense one form of energy input and output in a different form of energy.
- Sensors can also be classified as passive or active. In passive sensors, the power required to produce the output is provided by the sensed physical phenomenon itself whereas the active sensors require external power source.
- In embedded system, sensor and actuators are used for controlling the system. Sensors are connected to input port. Actuators are connected to output port.
- Sensor captures the changes in the environmental variable. Middle system process the information. Actuators are changed according to the input variable. It displays the output.
- Example of control is air conditioner system. It controls the room temperature to a specified limit.
- Deflection : The signal produces some physical (deflection) effect closely related to the measured quantity and transuded to be observable.
- Null : The signal produced by the sensor is counteracted to minimize the deflection. That opposing effect necessary to maintain a zero deflection should be proportional to the signal of the measurand.
- Here, the output is usually an 'electrical quantity' and measurand is a 'physical quantity, property or condition which is to be measured'.

- A sensor is a device that responds to a physical stimulus, measures the physical stimulus quantity and converts it into a signal usually electrical, which can be read by an observer or by an instrument.
- A sensor can be very small and itself can be a trackable devices. The sensor itself, if not connected is not part of the IoT or WSN value chain.

❑ Specifications of Sensor :

1. **Accuracy** : Error between the result of a measurement and the true value being measured.
2. **Resolution** : The smallest increment of measure that a device can make.
3. **Sensitivity** : The ratio between the change in the output signal to a small change in input physical signal. Slope of the input-output fit line.
4. **Repeatability/Precision** : The ability of the sensor to output the same value for the same input over a number of trials.
5. **Bandwidth** : The frequency range between the lower and upper cut-off frequencies, within which the sensor transfer function is constant gain or linear.

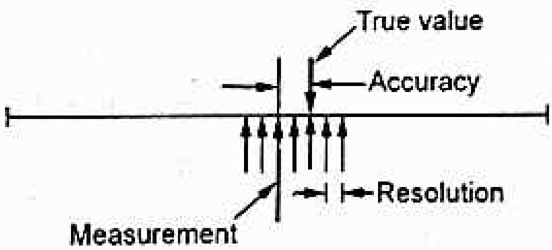

Fig. 2.6.1 : Accuracy vs. Resolution

2.6.1 Types of Sensors

- Mechanical sensor : Any suitable mechanical / electrical switch may be adopted but because a certain amount of force is required to operate a mechanical switch it is common to use micro-switches.
- Pneumatic sensor : These proximity sensors operate by breaking or disturbing an air flow. The pneumatic proximity sensor is an example or a contact type sensor. These cannot be used where light components may be blown away.
- Optical sensor : In there simplest form, optimcal proximity sensors operate by breaking a light beam which falls onto a light sensitive device such as a photocell. These are examples of non contact sensors. Care must be exercised with the lighting environment of these sensors for example optical sensors can be blinded by flashes from are welding processes, airborne dust and smoke clouds may impede light transmission etc.

- Electrical sensor : Electrical proximity sensors may be contact or non-contact. Simple contact sensors operate by making the sensor and the component complete an electrical circuit. Non-contact electrical proximity sensors rely on the electrical principles of either induction for detecting metals or capacitance for detecting non metals as well.
- Range sensing : Range sensing concerns detecting how near or far a component is from the sensing position, although they can also be used as proximity sensors. Distance or range sensors use non-contact analog techniques. Short range sensing, between a few millimetres and a few hundred millimetres is carried out using transmitted energy waves of various types e.g. radio waves, sound waves and lasers.

2.6.2 Sensor Data Communication Protocols

1. Direct transmission protocols :

- In direct communication protocol, each sensor sends its data directly to the base station. If the base station is far away from the nodes, direct communication will require a large amount of transmit power from each node.
- This will quickly drain the battery of the nodes and reduce the system lifetime. However, the only receptions in this protocol occur at the base station, so if either the base station is close to the nodes, or the energy required receiving data is large, this may be an acceptable method of communication.

2. Minimum transfer energy protocols :

- In these protocols, nodes act as routers for other nodes' data in addition to sensing the environment. These protocols differ in the way the routes are chosen.
- Some of these protocols only consider the energy of the transmitter and neglect the energy dissipation of the receivers in determining the routes.
- Depending on the real time costs of the transmit amplifier and the radio electronics, the total energy expended in the system might actually be greater using MTE routing than direct transmission to the base station.
- It is clear that in MTE routing, the nodes closest to the base station will be used to route a large number of data messages to the base station.
- Thus, these nodes will die out quickly, causing the energy required to get the remaining data to the base station to increase and more nodes to die.

2.6.3 Actuators

- A device or mechanism capable of performing a physical action. Actuators interact with the world. Sensors capture information from the world.
- The interface between the microcontroller and the sensors or the actuators is either analog or digital.
- An actuator requires a control signal and a source of energy. An actuator is the mechanism by which a control system acts upon an environment. The control system can be simple, software-based, a human, or any other input.
- When the actuation is a motion, motor have to be used for rotational or linear motion.

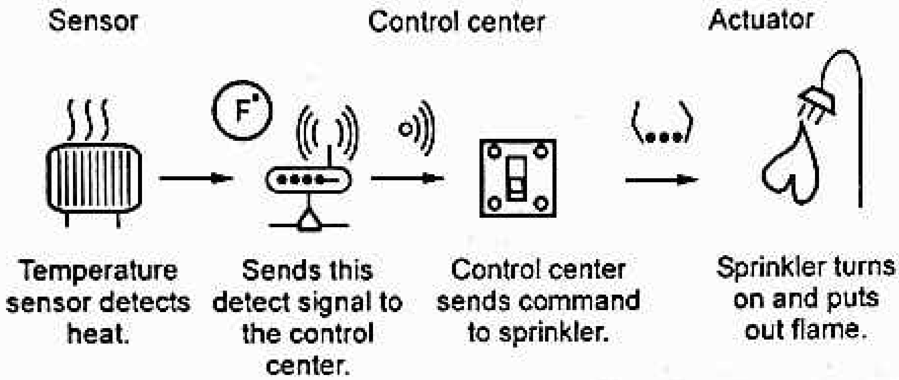

Fig. 2.6.2

- The selection of the proper actuator is more complicated than selection of the sensors, primarily due to their effect on the dynamic behaviour of the overall system. Furthermore, the selection of the actuator dominates the power needs and the coupling mechanisms of the entire system.
- In typical IoT systems, a sensor may collect information and route to a control centre where a decision is made and a corresponding command is sent back to an actuator in response to that sensed input.

2.6.4 Need of Analog / Digital Conversion

- An analog to digital converter (ADC) converts an analog signal into digital form. An embedded system uses the ADC to collect information about the external world (data acquisition system.)
- The input signal is usually an analog voltage, and the output is a binary number. The ADC precision is the number of distinguishable ADC inputs.
- By converting an analog signal to digital, we are effectively approximating it, as any one digital output value has to represent a very small range of analog input voltages, i.e. the width of any of the steps on the "staircase" n.

Examples of A/D Applications

1. **Microphones -** Take your voice varying pressure waves in the air and convert them into varying electrical signals
2. **Strain Gages -** Determines the amount of strain (change in dimensions) when a stress is applied
3. **Thermocouple** - Temperature measuring device converts thermal energy to electric energy
4. Voltmeters
5. Digital Multimeters.

- A digital to analog converter (DAC) converters a digital input represented as a binary number to an analogue voltage (or current) that is proportional to the value of this input.
- In general, the input quantity to the DAC is a digital number, the conversion techniques convert the number into corresponding number of units of current, voltage or charge and then these units are added together with an analog summing circuits.

2.7 Logical Design of IoT

- It is an abstract representation of an entities and processes but it can not specify low level implementation.

2.7.1 IoT Functional Blocks

- The Functional Model (FM) is derived from internal and external requirements. Functional view is derived from the Functional Model in conjunction with high-level requirements.
- IoT Functional model identifies Functional Groups (FGs) that is, groups of functionalities, grounded in key concepts of the IoT Domain Model.
- Functional Model is an abstract framework for understanding the main Functionality Groups (FG) and their interactions. This framework defines the common semantics of the main functionalities and will be used for the development of IoT-A compliant Functional Views.
- The Functional Model is not directly tied to a certain technology, application domain, or implementation. It does not explain what the different Functional Components are that make up a certain Functionality Group.

- Fig. 2.7.1 shows IoT functional model.
- The Application, Virtual Entity, IoT Service, and Device FGs are generated by starting from the User, Virtual Entity, Resource, Service, and Device classes from the IoT Domain Model.
- Device functional group contains all the possible functionality hosted by the physical devices. Device functionality includes sensing, actuation, processing, storage, and identification components, the sophistication of which depends on the device capabilities.
- Communication functional group support all the communication used by devices. It uses wired and wireless technology.

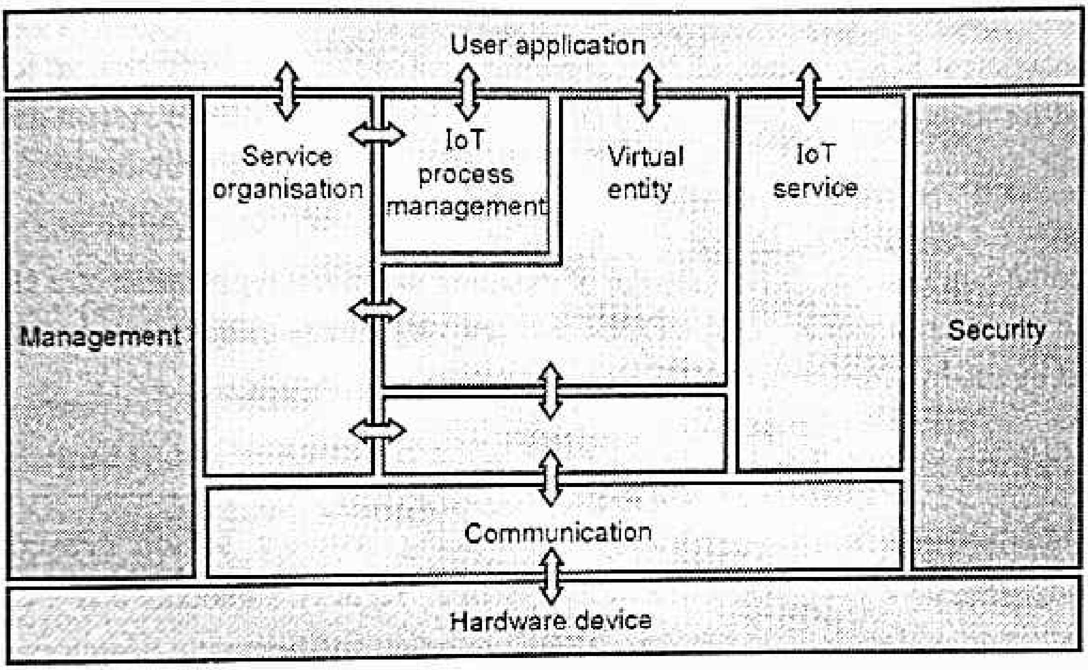

Fig. 2.7.1 : IoT functional model

- IoT Service functional group : Support functions such as directory services, which allow discovery of Services and resolution to resources.
- Virtual Entity functional group : It is related to the Virtual Entity class in the IoT Domain Model. Associations between Virtual Entities can be static or dynamic depending on the mobility of the Physical Entities related to the corresponding Virtual Entities.
- IoT Service Organization functional group : To host all functional components that support the composition and orchestration of IoT and Virtual Entity services.
- Finally, the "Management" transversal FG is required for the management of and/or interaction between the functionality groups.
- The IoT Process Management FG relates to the conceptual integration of (business) process management systems with the IoT ARM.

- The Service Organisation FG is a central Functionality Group that acts as a communication hub between several other Functionality Groups.
- The Virtual Entity and IoT Service FGs include functions that relate to interactions on the Virtual Entity and IoT Service abstraction levels, respectively.
- The Virtual Entity FG contains functions for interacting with the IoT System on the basis of VEs, as well as functionalities for discovering and looking up Services that can provide information about VEs, or which allow the interaction with VEs.
- Communication FG provides a simple interface for instantiating and for managing high-level information flow. It can be customized according to the different requirements defined in the Unified Requirements list.
- The Management FG combines all functionalities that are needed to govern an IoT system. The need for management can be traced back to at least four high-level system goals : Cost reduction; Attending unexpected usage issues; Fault handling and Flexibility.
- The Security Functionality Group is responsible for ensuring the security and privacy of IoT-A-compliant systems. It is in charge of handling the initial registration of a client to the system in a secure manner. This ensures that only legitimate clients may access services provided by the IoT infrastructure.

2.8 IoT Enabling Technologies

- IoT is enabled by several technologies including wireless sensor networks, cloud computing, Big data analytics, Embedded Systems, Security Protocols and architectures, communication protocols, web services, Mobile Internet, and Semantic Search engines.

2.8.1 Cloud Computing

- Cloud computing has the almost unlimited capacity of storage and processing power which is a more mature technology at least to a certain extent to solve the problem of most of the Internet of things.
- Cloud computing is a pay-per-use model for enabling available, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, services) that can be rapidly provisioned and released with minimal management effort or service-provider interaction.
- Cloud storage services may be accessed through a web service API, a cloud storage gateway or through a web-based user interface.
- **Cloud computing services are offered to users in different forms :** Infrastructure- as-a-Service (IaaS), Platform-as-a-Service (PaaS) and Software-as-a-Service (SaaS).

- **Software as a Service (SaaS) :** Model in which an application is hosted as a service to customers who access it via the Internet. The provider does all the patching and upgrades as well as keeping the infrastructure running. The traditional model of software distribution, in which software is purchased for and installed on personal computers.
- **Platform as a Service (PaaS) :** Platform as a service is another application delivery model and also known as cloud-ware. Supplies all the resources required to build applications and services completely from the Internet, without having to download or install software. Services includes application design, development, testing, deployment, and hosting, team collaboration, web service integration, database integration, security, scalability, storage, state management, and versioning. PaaS is closely related to SaaS but delivers a platform from which to work rather than an application to work with.
- **Infrastructure as a Service (IaaS) :** SaaS and PaaS are providing apples to customers, IaaS doesn't. It offers the hardware so that your organization can put whatever they want onto it. Rather than purchase servers, software, racks, and having to pay for the datacenter space for them, the service provider rents for resources like server space, network equipment, memory etc.

2.8.2 Big Data Analytic

- A category of technologies and services where the capabilities provided to collect, store, search, share, analyze and visualize data which have the characteristics of high-volume, high-velocity and high-variety.
- Examples of big data generated by IoT systems :
  a) Weather Monitoring Stations
  b) Machine sensor data from industrial systems
  c) Health and fitness data
  d) Location and tracking systems

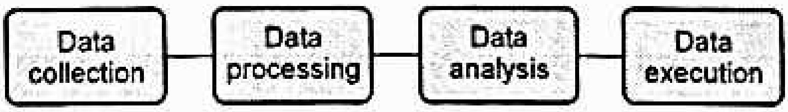

Fig. 2.8.1

2.8.3 Wireless Sensor Networks

- A Wireless Sensor Network (WSN) is a network formed by a large number of sensor nodes where each node is equipped with a sensor to detect physical phenomena such as light, heat, pressure, etc.

- WSNs nowadays usually include sensor nodes, actuator nodes, gateways and clients. A large number of sensor nodes deployed randomly inside of or near the monitoring area, form networks through self-organization.
- Sensor nodes monitor the collected data to transmit along to other sensor nodes by hopping. During the process of transmission, monitored data may be handled by multiple nodes to get to gateway node after multi-hop routing, and finally reach the management node through the internet or satellite.
- Standards for WSN technology have been well developed, such as Zigbee (IEEE 802.15.4). The IEEE 802.15.4 is simple packet data protocol for lightweight wireless networks.
- It works well for long battery life, selectable latency for controllers, sensors, remote monitoring and portable electronics.

2.8.4 Communication Protocols

- Communication protocols are used as a backbone of IoT systems. It enable network connectivity and coupling to applications. Communication protocols allow devices to exchange data over the network.
- Communication protocol also performs error correction and detection, flow control, data encoding, addressing mechanism etc.
- Sequence control, lost of packet, retransmission are the other functions of communication protocol.

2.8.5 Embedded System

- A system is a set of interacting or interdependent component parts forming a complex unit. It is a fixed plan to perform one or many task.
- Embedded system is an electronic system which is designed to perform one or a limited set of functions using software and hardware.
- General definition of embedded systems is : embedded systems are computing systems with tightly coupled hardware and software integration that are designed to perform a dedicated function. The word embedded reflects the fact that these systems are usually an integral part of a larger system, known as the embedding system. Multiple embedded systems can coexist in an embedding system.
- An embedded system has three main components: Hardware, Software and time operating system.

- Hardware parts includes power supply, processor, memory, times & communication ports, system application specific circuit etc.
- Software parts includes the application software is required to perform the series of tasks. An embedded system has software designed to keep in view of three constraints :
  a. Availability of System Memory
  b. Availability of processor speed
  c. The need to limit power dissipation when running the system continuously incycles of wait for events, run , stop and wake up.
- Demand for low cost and higher density platform requires the integration of devices. As integration levels increases, more and more logic is added to the processor die, creating families of applications specific service processors.
- System-on-Chip (SoC) designs increasingly become the driving force of a number of modern electronics systems. A number of key technologies integrate together in forming the highly complex embedded platform.

2.9 IoT Levels and Deployment Templates

- IoT system consists of following components
  1 Device : IoT device performs identification, remote monitoring, sensing and actuating functions.
  2 Resource : IoT device used software components as resources for accessing, processing, and storing sensor information. It also controls actuators.
  3 Controller Service : It sends data from the device to the web service and receives commands from the application for controlling the device.
  4 Database : IoT generates large amount of local database and cloud database.
  5 Web Service : Web services act as a link between the IoT device, application, database and analysis components.
  6 Analysis Component : It is responsible for analysing the IoT data and generate results.
  7. Application : User use this interface for controlling and monitoring various IoT system.
- Levels of IoT system are IoT Level-1, IoT Level-2, IoT Level-3, IoT Level-4, IoT Level-5, IoT Level-6.

- IoT Development level are six types.

IoT Level 1	Single node, perform sensing, perform analysis and hosts the application
IoT Level 2	Single node, perform sensing, perform local analysis
IoT Level 3	Single node, data is analyzed in cloud and application is cloud based
IoT Level 4	Multiple Node, perform local analysis, application is cloud based
IoT Level 5	Multiple end node and coordinator node,
IoT Level 6	Multiple independent node, perform sensing, send data to cloud

2.9.1 IoT Level 1

- Physical devices and controllers that might control multiple devices. These are the "things" in the IoT, and they include a wide range of endpoint devices that send and receive information.
- Fig. 2.9.1 shows IoT level 1.

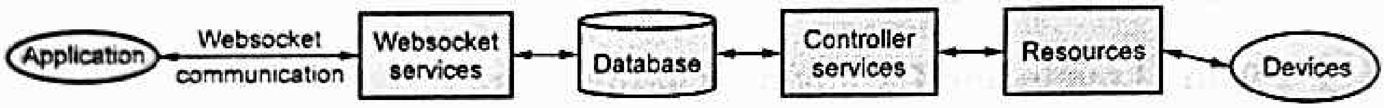

Fig. 2.9.1 : IoT level 1

- Level 1 IoT systems are suitable for modeling low-cost and low complexity solutions where the data involved is not big and the analysis requirements are not computationally intensive.
- Eg : Home automation
- The system consists of a single node that allows controlling the lights and appliances in a home remotely. The device used in the system interfaces with the light and appliances using electronic relay switches. The status information of each light or appliance is maintained in the local database. The controller service continuously monitors the state of each light or appliance and triggers the relay switches accordingly.

2.9.2 IoT Level 2

- In level 2, single node performs sensing and local analysis. Fig. 2.9.2 shows block diagram.

Fig. 2.9.2

- Both data and application is stored on the cloud. This type of system is suitable for big data used for analysis and that to performed on local side.
- Smart irrigation is an example of level 2 IoT system. System uses single sensor for monitoring the soil moisture level and controls the irrigation system.
- Controller service monitors continuously the moisture level. If the moisture level drops below a threshold, the irrigation system is turned ON.

2.9.3 IoT Level 3

- Single node is used in level 3 internet of things. Collected data is stored on the cloud and processed on the cloud. Application is also stored on the cloud.
- This is suitable for huge data and analysis requirement are computationally intensive.

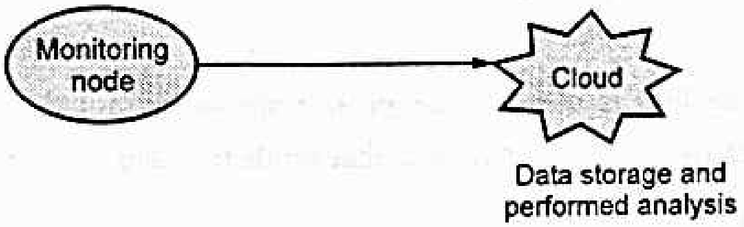

Fig. 2.9.3

- REST or websocket is used for communicating with client machine. Consider an example of tracking package handling.
- Single node monitors the vibration level for the package being shipped. For monitoring vibration level, accelerometer and gyroscope sensor is used.
- Accelerometer and gyroscope sensor is used. System allows shippers tracking cargo to communicate with the devices, changing reporting frequencies and investigating alerts generated by attached sensors.
- Controller device sends the sensor data to the cloud in real time using WebSocket service. Data is stored on the cloud and analysis is also performed on the cloud.
- Cloud can send alerts if the vibration levels is greater than threshold level.

2.9.4 IoT Level 4

- Level 4 IoT system uses multiple nodes (sensors) and processing is performed on local node. Data is stored on the cloud and application is stored on the cloud storage.
- Fig. 2.9.4 shows block diagram of Level 4 IoT.

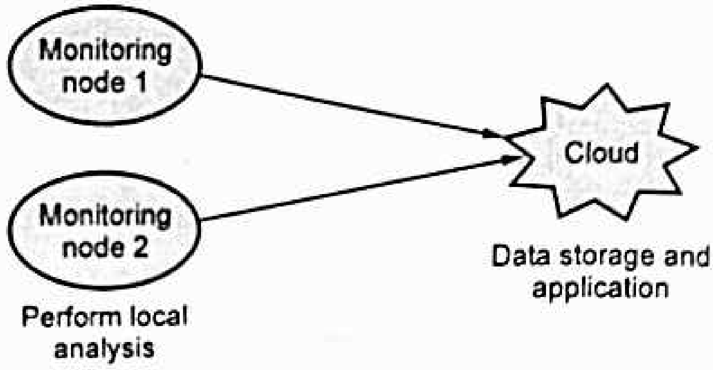

Fig. 2.9.4

- Level 4 IoT uses two observer sensor for local and cloud. These sensor subscribe and receive information form the IoT device to cloud.
- The observer sensor can process information and used for various purposes. Control function is not performed by this observer sensor.
- Noise monitoring system is used IoT level 4. Sensors are not depends upon each other. Each sensor runs its own controller service that sends the data to the cloud.

2.9.5 IoT Level 5

- It contains multiple end sensor and one coordinator sensor. Fig. 2.9.5 shows IoT level 5 block diagram.
- The end sensor perform sensing and/or actuation. Coordinator sensor collect data from the end sensor and sends to the cloud.
- Data is stored and analysis in the cloud and application is also cloud based. Forest fire detection system uses level 5 IoT system.

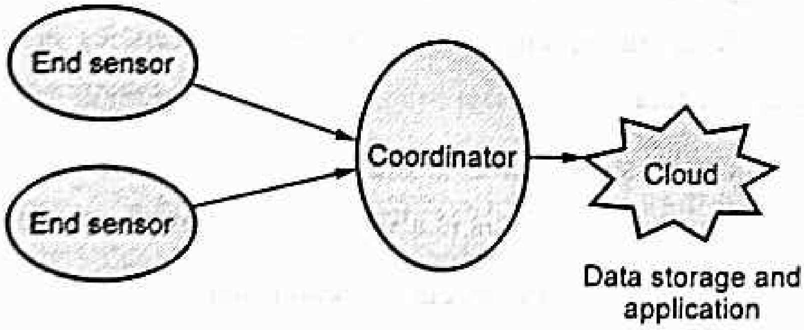

Fig. 2.9.5

- Multiple sensor are kept at different location to monitor temperature, humidity, carbon dioxide level. Coordinator node (sensor) collect data for end node and these nodes act as gateway and provides Internet services to the IoT systems.

2.9.6 IoT Level 6

- It contains multiple independent end nodes and it performs sensing and/or actuation function. It sends data to the cloud. Fig. 2.9.6 shows block diagram of level 6 IoT.
- Data is stored on the cloud and application is also cloud based. Result is displayed on the cloud. Centralized controller knows the status of monitoring node and sends control commands to the nodes.
- Weather monitoring system uses Level 2 IoT based system. Multiple nodes are kept at different location to monitor temperature, humidity level.

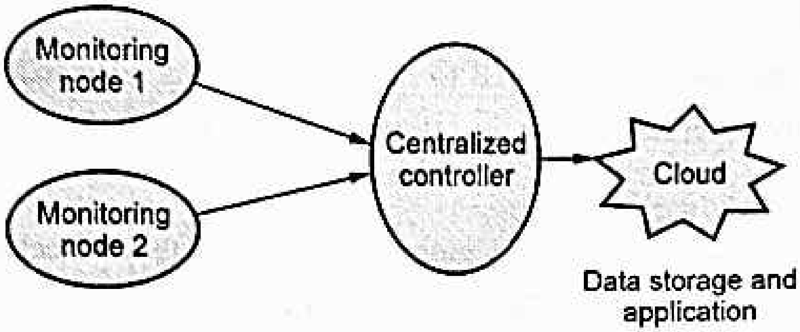

Fig. 2.9.6

- End node send real time data to cloud using WebSocket service. Cloud database is used for storing data. Data analysis is performed on cloud side. Cloud based application is used for display data.

2.10 Challenges for IoT

- IoT and ubiquitous integration of clinical environments define complex design challenges and requirements in order to reach a suitable technology maturity for its wide deployment and market integration.
- From the beginning, IoT devices present inherited challenges since they are constrained devices with low memory, processing, communication and energy capabilities.

1. The first key challenge for a ubiquitous deployment is the integration of multi-technology networks in a common all-IP network to ensure that the communication network is reliable and scalable. For this purpose, IoT relies on the connectivity and reliability for its communications on Future Internet architecture.
2. The second key challenge is to guarantee security, privacy, integrity of information and user confidentiality. The majority of the IoT applications need to take into considerations the support of mechanisms to carry out the authentication, authorization, access control, and key management.

3. In addition, due to the reduced capabilities from the constrained devices enabled with Internet connectivity, a higher protection of the edge networks needs to be considered with respect to the global network.

4. The third key challenge is to offer support for the mobility, since the Future Internet presents a more ubiquitous and mobile Internet. Mobility support increases the applicability of Internet to new areas.

- The most present nowadays are mobile platforms such as smart phones and tablets which enable a tremendous range of applications based on ubiquitous location, context awareness, social networking, and interaction with the environment.
- Future Internet potential is not limited to mobile platforms, else IoT is another emerging area of the Future Internet, which is offering a high integration of the cybernetic and physical world.
- Mobility support in the IoT enables a global and continuous connection of all the devices without requiring the disruption of the communication sessions.
- For example, mobility management in hospitals is required since clinical devices can be connected through wireless technologies. Mobility offers highly valuable features such as higher quality of experiences for the patients, since this allows the patients to move freely, continuous monitoring through portable/wearable sensors, extend the coverage within all the hospital, and finally a higher fault tolerance since the mobility management allows the connection to adapt dynamically to different access points.
- Therefore, clinical environment is one of the main scenarios where the mobility for the IoT-based applications exploit these capabilities, in terms of fault tolerance influences directly in the life support, and continuous monitoring influences the quantity of data available which is required for real-time diagnostic.

5. Other challenges are also arising from the application, economical, and technological perspectives. For example, from an application point of view are the requirements for processing large amounts of data for a growing number of devices, it is called Big Data.

- From the economic points of view, the needs to provide economies of scale, i.e., new services based on existing modules in order to leverage the related platform investment.
- From the networking point of view to offer an end-to-end support for Quality of Service , since the different IoT applications will present different requirements in terms of latency and bandwidth, for example, for clinical environments the traffic should be prioritized over other non-critical traffic coming from smart-metering.

- Fig. 2.10.1 shows the key challenges to offer an Internet of Everything. This covers from the integration of heterogeneous devices to the integration into a Web of Things.

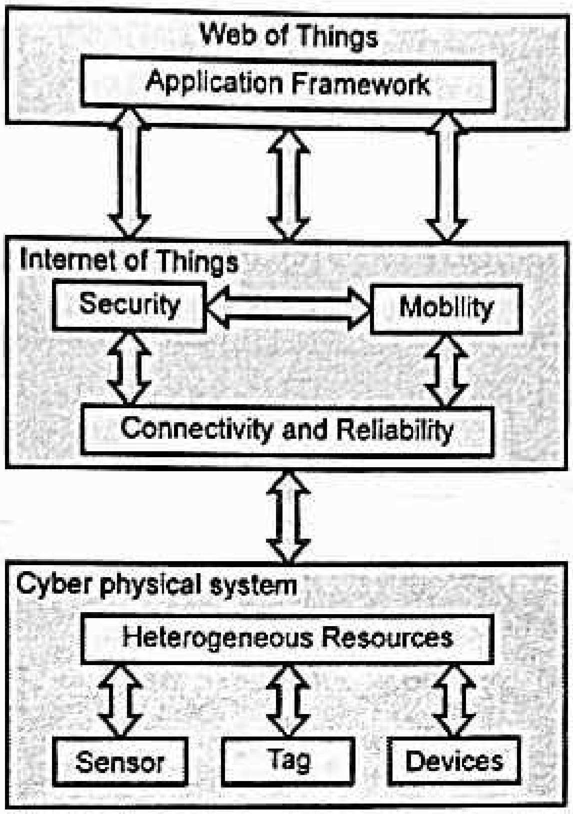

Fig. 2.10.1 : Key challenges to offer an Internet of everything

- The number of devices that are connected to the Internet is growing exponentially. This has led to defining a new conception of Internet, the commonly called Future Internet, which started with a new version of the Internet Protocol (IPv6) that extends the addressing space in order to support all the emerging Internet-enabled devices.
- IoT devices are small wireless devices that would be placed in public places. Wireless communication is made secure through encryption technique. But the IoT devices are very small and not powerful enough to support encryption methods. There is need to modify encryption algorithm in order to support IoT devices.
- In IoT, the different devices are traceable through the interconnected network, it creates threats to personal and private data.
- Many IoT devices are small in size and do not have the continuous power source. A device computation depends on battery size and cost of the device. Many IoT devices work as a single, limited purpose which could have customized network interfaces, operating systems, and programming models that make the most efficient use of limited computation, network, and energy resources.